

0 1 . 1

In the following passage, the numbered spaces can be filled with biological terms.

Glycolysis is the first stage of respiration. It occurs in the ____ (1) ____ of cells and is an anaerobic process. Glycolysis involves the ____ (2) ____ of glucose to glucose phosphate, using ATP. The oxidation of triose phosphate to ____ (3) ____ results in the net gain of ATP and reduced ____ (4) ____.

Write the correct biological term beside each number below that matches the space in the passage.

[2 marks]

1 _____

2 _____

3 _____

4 _____

0 1 . 2

The equation for the aerobic respiration of glucose is:



Six carbon dioxide molecules are produced from each glucose molecule.

Use your knowledge of the link reaction **and** Krebs cycle to explain why.

[2 marks]

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An inhibitor of the electron transfer chain was added to aerobically respiring yeast cells.

As aerobic respiration was inhibited, the following changes occurred:

- oxygen uptake decreased
- ATP production decreased
- ethanol was produced.

Explain the changes in the yeast cells following the addition of the inhibitor.

[3 marks]

Oxygen uptake decreased _____

ATP production decreased _____

Ethanol was produced _____

Turn over for the next question

